

Assignment: Loan Default Prediction Using Machine Learning

Objective

In this assignment, you will build a **Machine Learning classification model** using the provided finance dataset.

Your goal is to **predict loan default risk** using customer financial attributes.

This assignment will help you understand:

- End-to-end ML workflow
- Data preprocessing
- Model training and evaluation

Dataset Description

The dataset contains customer financial indicators used to predict loan default.

Column	Description
customer_id	Unique identifier for each customer
age	Age of the customer
annual_income	Annual income
credit_score	Credit score of customer
loan_amount	Loan amount requested
existing_loans	Number of existing loans
loan_default	Target variable (0 = No Default, 1 = Default)

Assignment Tasks

You must complete the following stages.

Part 1 – Data Understanding & Exploration

Perform basic **data exploration and preprocessing**.

Tasks

1. Load the dataset using **Pandas**
2. Explore dataset structure:
 - Shape of dataset
 - Column types
 - Missing values
3. Perform **basic statistical analysis**
4. Visualize relationships between features and loan default.

Suggested Visualizations

- Age vs Loan Default
- Credit Score Distribution
- Annual Income Distribution
- Loan Amount vs Loan Default
- Correlation Heatmap

Deliverables

- Key insights from dataset
- Observations about potential predictors

Part 2 – Machine Learning Classification Model

Build a **Machine Learning model** to predict loan default.

Tasks

1. Separate **features (X)** and **target variable (y)**
2. Perform **train-test split**
3. Apply **feature scaling**
4. Train a **Logistic Regression classifier**

5. Generate predictions
6. Evaluate model performance

Evaluation Metrics Required

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1 Score

Deliverables

- Model accuracy
- Confusion matrix interpretation

Expected Learning Outcomes

Outcome 1

Perform **data preprocessing and exploratory analysis** on financial data.

Outcome 2

Build a **complete machine learning classification pipeline** including training, prediction, and evaluation.

Outcome 3

Evaluate model performance using **classification metrics and confusion matrices**.